Supporting Materials for "A Modified Labeling-Oriented Non-Dominated Sorting Algorithm"

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1 Supporting Materials for LONSA

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\begin{aligned}
A_L &= \sum_{l=1}^{K}\sum_{k=l}^{K}(n_k-1)\left(n_{k+1}+n_{k+2}+\cdots+n_K\right) \\
&= \underbrace{\sum_{l=1}^{K}\sum_{k=l}^{K} n_k\left(n_{k+1}+n_{k+2}+\cdots+n_K\right)}_{X_L} - \underbrace{\sum_{l=1}^{K}\sum_{k=l}^{K}\left(n_{k+1}+n_{k+2}+\cdots+n_K\right)}_{Y_L} \\
&= \sum_{k=1}^{K-1} k n_k\left(n_{k+1}+n_{k+2}+\cdots+n_K\right) - \frac{1}{2}\sum_{k=2}^{K} k(k-1)n_k \quad \text{From Equation (2), (3)}
\end{aligned} \tag{1}
$$

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\begin{aligned}
X_L &= \sum_{l=1}^{K}\sum_{k=l}^{K} n_k\left(n_{k+1}+n_{k+2}+\cdots+n_K\right) \\
&= \left[n_1\left(n_2+n_3+\cdots+n_K\right)+n_2\left(n_3+n_4+\cdots+n_K\right)+\cdots+n_{K-1}\left(n_K\right)\right]+ \\
&\quad \left[n_2\left(n_3+n_4+\cdots+n_K\right)+n_3\left(n_4+n_5+\cdots+n_K\right)+\cdots+n_{K-1}\left(n_K\right)\right]+ \\
&\quad \left[n_3\left(n_4+n_5+\cdots+n_K\right)+n_4\left(n_5+n_6+\cdots+n_K\right)+\cdots+n_{K-1}\left(n_K\right)\right]+ \\
&\quad \cdots+\left[n_{K-1}\left(n_K\right)\right] \\
&= n_1\left(n_2+n_3+\cdots+n_K\right)+2n_2\left(n_3+n_4+\cdots+n_K\right)+\cdots+(K-1)n_{K-1}\left(n_K\right) \\
&= \sum_{k=1}^{K-1} k n_k\left(n_{k+1}+n_{k+2}+\cdots+n_K\right)
\end{aligned} \tag{2}
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\begin{aligned}
Y_L &= \sum_{l=1}^{K}\sum_{k=l}^{K}(n_{k+1}+n_{k+2}+\cdots+n_K) \\
&= [(n_2+n_3+\cdots+n_K)+(n_3+n_4+\cdots+n_K)+\cdots+(n_K)]+ \\
&\quad [(n_3+n_4+\cdots+n_K)+(n_4+n_5+\cdots+n_K)+\cdots+(n_K)]+ \\
&\quad [(n_4+n_5+\cdots+n_K)+(n_5+n_6+\cdots+n_K)+\cdots+(n_K)]+\cdots+ \\
&\quad [(n_K)] \\
&= [n_2+2n_3+\cdots+(K-1)n_K]+[n_3+2n_4+\cdots+(K-2)n_K]+ \\
&\quad [n_4+2n_5+\cdots+(K-3)n_K]+\cdots+[(n_K)] \\
&= n_2+(2+1)\,n_3+(3+2+1)\,n_4+(4+3+2+1)\,n_5+\cdots+ \\
&\quad ((K-1)+(K-2)+\cdots+1)\,n_K \\
&= \sum_{k=2}^{K}[(k-1)+(k-2)+\cdots+1]\,n_k \\
&= \frac{1}{2}\sum_{k=2}^{K}k(k-1)n_k \qquad (3)
\end{aligned}
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\begin{aligned}
B_L &= \frac{1}{2}\sum_{l=1}^{K}\sum_{k=l}^{K}n_k(n_k-1) \\
&= \frac{1}{2}\left[\sum_{l=1}^{K}\sum_{k=l}^{K}(n_k^2-n_k)\right] \\
&= \frac{1}{2}\left[\sum_{l=1}^{K}\sum_{k=l}^{K}n_k^2\right]-\frac{1}{2}\left[\sum_{l=1}^{K}\sum_{k=l}^{K}n_k\right] \\
&= \frac{1}{2}\left[\left(n_1^2+n_2^2+\cdots+n_K^2\right)+\left(n_2^2+n_3^2+\cdots+n_K^2\right)+\cdots+\left(n_K^2\right)\right]- \\
&\quad \frac{1}{2}\left[(n_1+n_2+\cdots+n_K)+(n_2+n_3+\cdots+n_K)+\cdots+(n_K)\right] \\
&= \frac{1}{2}\left(n_1^2+2n_2^2+3n_3^2+\cdots+Kn_K^2\right)-\frac{1}{2}\left(n_1+2n_2+3n_3+\cdots+Kn_K\right) \\
&= \left[\frac{1}{2}\sum_{k=1}^{K}kn_k^2\right]-\left[\frac{1}{2}\sum_{k=1}^{K}kn_k\right] \\
&= \frac{1}{2}\sum_{k=1}^{K}kn_k(n_k-1) \qquad (4)
\end{aligned}
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\begin{aligned}
C_L &= \sum_{l=1}^{K}\sum_{k=l+1}^{K}n_k \\
&= (n_2+n_3+\cdots+n_K)+(n_3+n_4+\cdots+n_K)+\cdots+(n_K) \\
&= n_2+2n_3+3n_4+\cdots+(K-1)n_K \\
&= \sum_{k=2}^{K}(k-1)n_k \qquad (5)
\end{aligned}
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\begin{aligned}
\mathtt{dC} &= \sum_{k=1}^{K-1} kZ(K-k)Z - \frac{1}{2}\sum_{k=2}^{K}(k-1)(k-2)Z + \frac{1}{2}\sum_{k=1}^{K} kZ(Z-1) \\
&= Z^2\sum_{k=1}^{K-1} k(K-k) - \frac{1}{2}Z\sum_{k=1}^{K}(k^2-3k+2) + \frac{1}{2}Z(Z-1)\sum_{k=1}^{K} k \\
&= Z^2K\sum_{k=1}^{K-1} k - Z^2\sum_{k=1}^{K-1} k^2 - \frac{1}{2}Z\sum_{k=2}^{K} k^2 + \frac{3}{2}Z\sum_{k=1}^{K} k - Z\sum_{k=1}^{K} 1 + \frac{1}{4}N(Z-1)(K+1) \\
&= NZ\frac{1}{2}K(K-1) - Z^2\frac{1}{6}(K-1)K(2K-1) - \frac{1}{2}Z\frac{1}{6}K(K+1)(2K+1) + \\
&\quad \frac{3}{2}Z\frac{1}{2}K(K+1) - ZK + \frac{1}{4}N(Z-1)(K+1) \\
&= \frac{1}{2}N^2(K-1) - \frac{1}{6}NZ(K-1)(2K-1) - \frac{1}{12}N(K+1)(2K+1) + \\
&\quad \frac{3}{4}N(K+1) - N + \frac{1}{4}N(KZ+Z-K-1) \\
&= \frac{1}{2}N^2(K-1) - \frac{1}{6}NZ(2K^2-3K+1) - \frac{1}{12}N(2K^2+3K+1) + \\
&\quad \frac{3}{4}N(K+1) - N + \frac{1}{4}N(N+Z-K-1) \\
&= \frac{1}{12}\left[6N^2K - 6N^2 - 4NZK^2 + 6NZK - 2NZ - 2NK^2 - 3KN - N + \right. \\
&\quad \left. 9NK + 9N - 12N + 3N^2 + 3NZ - 3NK - 3N\right] \\
&= \frac{1}{12}\left[6N^2K - 6N^2 - 4N^2K + 6N^2 - 2NZ - 2NK^2 - 3KN - N + \right. \\
&\quad \left. 9NK + 9N - 12N + 3N^2 + 3NZ - 3NK - 3N\right] \\
&= \frac{1}{12}\left[\left(6N^2K - 4N^2K\right) + (-2NZ + 3NZ) - 2NK^2 + \right. \\
&\quad \left. (-3KN - 3NK + 9NK) + (-N + 9N - 12N - 3N) + 3N^2\right] \\
&= \frac{1}{12}\left(2N^2K + NZ - 2NK^2 + 3NK - 7N + 3N^2\right) \\
&= \frac{1}{12}N\left(2NK - 2K^2 + 3K - 7 + 3N + Z\right) \qquad (6)
\end{aligned}
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